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# Acetylene black interlayer regulated sulfur deposition for lithium-sulfur batteries with high utilization and long-term life

Xin-Liang Men, Teng Deng, Xue-Chao Jiao, Lai-Tao Qu, Kai-Ning Wen, Jiang-Xuan Che, Li-Ping Chen, Sen Li, Juan Wang *

*School of Mechanical & Electrical Engineering, Xi'an Key Laboratory of Clean Energy, Shaanxi Key Laboratory of Nanomaterials and Nanotechnology, Xi'an University of Architecture and Technology, Xi'an 710055, PR China*

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ABSTRACT

Lithium-sulfur (Li-S) batteries are considered to be competitive next-generation energy storage systems. However, practical applications are limited due to the shuttle of lithium polysulfides (LiPSs) and the poor electronic conductivity of the sulfur cathode, especially at high-rate currents. Here, we applied commercialized Acetylene black (AB) as separator coating in Li-S batteries. The stacking of AB nanospheres forms a void space that can physically adsorb polysulfides, store electrolytes, and promote lithium-ion diffusion; Simultaneously, it forms an excellent electronic conduction network that significantly improves the utilization of higher-order LiPSs. The thickness and composition of the coating can regulate the nucleation of crystalline S and $Li_2S$ to promote the uniform deposition of solid-phase particles. AB-150 releases an initial capacity of 961.3 mAh $g^{-1}$ and still maintains at 825 mAh $g^{-1}$ after 100 cycles with a capacity retention rate of 85.8% at 0.2C. Even remains at 371.2 mAh $g^{-1}$ after 1000 cycles at 1C with a weak capacity decay rate of 0.06% per cycle. These understandings can also be extended to the design of polar/catalytic interlayer materials driving the large-scale application of Li-S batteries.

## 1. Introduction

Lithium-sulfur (Li-S) batteries are considered promising energy storage systems because of sulfur's high specific capacity (1672 mAh $g^{-1}$), low price, abundant reserves, and environmental friendliness [1–3]. However, there are still serious challenges to its practical application. For example, the electronic insulating properties of sulfur(S) and lithium sulfide($Li_2S$) and leads to severe polarization of the battery and even operation termination; the shuttling of soluble intermediates (lithium polysulfides: LiPSs) during charge and discharge leads to severe capacity decline [4–7]. Various strategies have been proposed to address these issues, such as designing sulfur cathode structures [8–12], interlayer, electrolyte to regulate solid phase transition [7,13–16], or separator surface modification [17–21]. Among them, most of the attempts focused on the design of sulfur cathodes, such as carbon-based porous materials [4,12,22,23]and metal compounds [24–30] to physically adsorb, chemically anchor, and catalyze LiPSs intermediates. Another effort has been devoted to adding new components, using vacuum filtration [31,32], or slurry coating [33,34] to modify the separator. Su et al. [35] used microporous carbon paper (≈1.5 nm) as an interlayer in Li-S batteries. The components not only provide an electron transfer path through the insulating S/$Li_2S$ to improve the utilization of active materials but also store dissolved LiPSs. Tian et al. [36] pointed out that the thickness of the AB/MWCTs coating directly affects the impedance and lithium-ion diffusion. Inspired by classical methods in the particle agglomeration process, Shaibani et al. [37] synthesized ultra-high viscosity C/S cathodes by tuning the ratio of CMC, C, and water, which release a capacity of ≈19 mAh $cm^{-2}$ at ultra-high sulfur loadings of 13 mg $cm^{-2}$. The authors pointed out that different blending methods and binder content can affect the free space between the cathode particles and binder, which can ensure the minimization of electrode impedance and ion diffusion in the case of high sulfur content cathodes. It is worth noting that the separator is made of glass fiber with a carbon coating that may also play a key role but it is not explored in the paper. the liquid-solid transition involves solid nucleation and growth as the rate-determining steps of the reaction [38], so deposition morphology of $Li_2S$ and S on the substrate is crucial for the continuous electrochemical reaction. For instance Cañas et al. [39] found sulfur crystallized on the

* Corresponding author.
*E-mail address:* juanwang@xauat.edu.cn (J. Wang).

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separator surface with a preferred orientation like "sulfur needles" aligned perpendicular. Nelson et al. [40] found that the recrystallization of sulfur cathode by the end of the charge cycle is dependent on the preparation technique of the sulfur. These studies demonstrate that the deposition morphologies of S and $Li_2S$ can be regulated by the structural design. Here, we utilize commercialized Acetylene black (AB) with high electronic conductivity and uniform particle size as a modified material of separator. AB coating forms a three-dimensional conductive framework and a large amount of free space to store electrolytes. During discharge, the LiPSs dissolved from the cathode spontaneously enter the AB coating under the action of the concentration gradient force and are subsequently anchored by the van der Waals force. The AB conductive layer is in contact with the positive electrode to transfer electrons to the external circuit so the LiPSs is reduced to solid $Li_2S$ into the AB adsorption layer, preventing the deactivation caused by the adsorption of the active material without reaction. The electrolyte stored in the AB coating effectively relieves the $Li^+$ diffusion limitation at a high-rated current thus enabling excellent rate performance. The loss of battery capacity is attributed to the blocking of the electron transfer in the local area induced by the insulating $Li_2S$ uneven deposited on the AB layer, resulting in the termination of the electrochemical reaction. Based on the above understanding, we improve the viscosity, component content, and coating thickness of the AB to regulate the uniform nucleation and deposition of S and $Li_2S$. The optimized AB interlayer Li-S batteries exhibit an initial discharge capacity of 934.6 mAh $g^{-1}$ and remain at 371.2 mAh $g^{-1}$ with a capacity decay rate of 0.06% per cycle in 1000 cycles at 1C.

## 2. Experimental

### 2.1. Preparation of CNTs/S cathode

Sulfur (S 70 wt %) and carbon nanotubes (CNTs 30 wt %) were ground for 30 min, calcined at 155 °C for 12 h under an $N_2$ atmosphere to obtain black fluffy blocks. CNTs/S, AB, and PVDF (mass ratio 7:2:1) were dispersed in NMP continuous stirring for 8h followed by a 200 um scraper to scrape it on the aluminum foil and then dry in a vacuum environment at 60 °C for 12 h, cutting into circular pieces (diameter=12 mm) with a sulfur loading of about 1.4-1.7 mg $cm^{-2}$. A high sulfur loading of 3.96 mg $cm^{-2}$ was prepared by adjusting the scraper height.

### 2.2. Preparation of AB Separator

AB (100 mg) was dispersed in 2.5 mL of Nitromethylpyrrolidone (NMP) containing 15 mg of Polyvinylidene fluoride(PVDF) continuous stirring for 8 h. A 150 μm scraper to scrape it on commercial Celgard 2500 and then dry in a vacuum environment at 60 °C for 12 h, cutting into circular pieces (diameter =19 mm) with an AB loading of about 1.0 mg $cm^{-2}$, marked as AB-150.

AB (100 mg) was dispersed in 1.3 mL of NMP containing 28.6 mg of PVDF continuous stirring for 8 h. Followed by a 200 μm scraper to scrape it on commercial Celgard 2500 and then stored in a vacuum environment at 60 °C for 12 h, cutting into circular pieces (diameter =19mm) with an AB loading of about 1.3 mg $cm^{-2}$, marked as AB-200.

### 2.3. $Li_2S_6$ symmetric cell assembly and evaluation of liquid-liquid conversion

The $Li_2S_6$ symmetry batteries were assembled using a CR2032 coin cell. In detail, two aluminum foils with a diameter of 12 mm were used as the positive and negative electrodes. 15 μL $Li_2S_6$ was added dropwise to AB functional separator with a diameter of 19 mm, and let it stand for 5 min to ensure that $Li_2S_6$ is immersed into the coating. The coated sides of the two AB separators immersed in $Li_2S_6$ face the positive and negative electrodes, respectively.

The evaluation of liquid-liquid conversion was performed using cyclic voltammetry with a voltage window of -1.5∼+1.5 V, a scan rate of 10 mV $s^{-1}$, and a data acquisition interval of 1s.

### 2.4. Evaluating liquid-solid conversion through $Li_2S$ precipitation and decomposition

$Li_2S$ precipitation and decomposition were tested on CR2032 coin cells. AB-coated aluminum foil, pp, and lithium metal were used as the positive, separator, and negative electrodes, respectively. The AB-coated aluminum foils were prepared the same as the corresponding separators except that the separators were replaced by aluminum foil. 15 μL of $Li_2S_8$ solution was added to the as-prepared AB electrode with a diameter of 16 mm, which was left to stand for 5 min to ensure that the $Li_2S_8$ solution was immersed in the AB coating. 15 μL of electrolyte was added to the side of the separator near the lithium metal to complete the battery assembly.

#### 2.4.1. $Li_2S$ precipitation test

The assembled cell was galvanostatic discharged at 0.113 mA to 2.06 V to ensure the conversion of $Li_2S_8$ to $Li_2S_4$. Then electrostatic discharge at 2.02 V promotes the conversion of liquid phase $Li_2S_4$ to solid phase $Li_2S$ until the current is less than 0.01mA.

#### 2.4.2. $Li_2S$ decomposition test

The assembled cell was galvanostatic discharged at 0.113 mA to 1.8 V, followed by constant voltage discharge at 1.8 V until the current was less than 0.01 mA to ensure complete conversion of polysulfides to solid phase $Li_2S$. It was subsequently charged at a constant voltage of 2.4 V until the current was less than 0.01 mA to ensure the decomposition of $Li_2S$.

### 2.5. Visualization of AB interlayer barrier $Li_2S_6$

Separate H-shaped glass bottles with pp or AB-150 or AB-200. Add 12 mL of $Li_2S_6$ electrolyte with a concentration of 4.16 mM to the glass bottle facing the AB coating side, and add the same volume of $Li_2S_6$-free electrolyte to the other side. Changes in solution color over time were recorded via a smartphone at the same time interval.

### 2.6. Material characterization

The crystal structure of CNTs/S was determined by X-ray diffraction (Japan Rigaku D/Max2550VB+/PC diffractometer equipped with Cu Kα radiation (λ = 1.5406 Å)) at a scan rate of 7° $min^{-1}$ in 10-70°. The microstructure and elements of the material were collected by Scanning electron microscopy (SEM, Hitachi S-4800, 5 keV) and Energy Dispersive Spectrometer (EDS) equipped with an energy dispersive X-ray analyzer. The sulfur content in the CNTs/S was determined by Thermogravimetric analysis (ISQ 7000 Nicolet iS50 TGA/ DSC3) under flowing $N_2$ at a heating rate of 10°C $min^{-1}$.

### 2.7. Electrochemical test

Li-S batteries were assembled using a CR2032 coin cell containing CNTs/S cathode, AB modified separator, Li anode, and 50 μL electrolytes (1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME, 1:1 v/v) with 1 M Li bis-trifluoromethanesulfonimide (LiTFSI) additive 2wt% $LiNO_3$) in an Ar-filled glove box.

Cyclic voltammetry curves were obtained over a potential window of 1.7-2.8 V vs. $Li/Li^+$ while electrochemical impedance spectroscopy (EIS) was collected over a frequency range of 0.01 Hz to 100,000 Hz by a CHI660C electrochemical work state. Charge-discharge information was collected using a galvanostatic charge-discharge mode with a voltage window of 1.7-2.8V while Galvanostatic intermittent titration technique (GITT) measurements were carried out under the current pulse and maintained for 20 min to obtain the closed-circuit voltages and then

subsequently rested for 20min to acquire the quasi-open-circuit-voltage at 0.1 C with a voltage window of 1.7-2.8V by the NEWARE-BTS-CT4008-5 V 10 mA system.

## 3. Results and discussion

Crystal information of CNTs/S composites was determined by X-ray diffraction pattern (XRD). As shown in Fig. 1a, the diffraction peaks located at 14, 23.08, and 27.76° are consistent with the orthogonal sulfur (PDF# 08-0247), corresponding to the (113), (222), (040) crystal planes, indicating a large amount of sulfur is combined with carbon nanotubes in heat treatment at 155 °C. The sulfur content of CNTs/S composite was determined to be 70 wt.%. As shown in Fig. 1b, the quality of CNTs/S hardly changes before 155 °C, drops rapidly between 155 °C and 330 °C because of the loss of sulfur transition to gaseous state, and remains unchanged from 330 °C to 600 °C corresponding to the content of CNTs. A digitized photograph of the AB-150 modified separator is shown in Fig. 1c with a length of 14 cm and a width of 7.5 cm.

The microstructure of the material was determined by scanning electron microscopy (SEM) image. As shown in Fig. 2a, the CNTs/S composites were clearly distinguishable tubular structures no bulk sulfur was found, indicating that S was homogeneously mixed with CNTs. After mixing CNTs/S and AB (Fig. 2b), the tubular was transformed into a spherical stacked structure, indicating that AB was well attached to the surface of CNTs/S, which was beneficial to enhancing the electronic conductivity of the composite. Commercial Celgard 2500 (Fig. 2c) has uniformly distributed channels with a pore length of approximately 300 nm. AB (Fig. 2d) with uniform particle size (diameter ≈ 50-70 nm). The morphology of the AB-modified interlayer was further explored, as shown in Fig. 2e and g. In AB-150 and AB-200, AB coating forms a large number of pores, which can store electrolyte and soluble LiPSs, corresponding coating thicknesses are 12.3 μm (Fig. 2e) and 21.87 μm (Fig. 2h), respectively.

Cyclic voltammetry curves were obtained to detect the electrochemical reaction process. As shown in Fig. 3a, all samples exhibit similar curves containing two reduction peaks and one oxidation peak. Peak C1, peakC2, and peakA correspond to the reaction of sulfur with lithium ions to form higher-order LiPSs ($Li_2S_x$ $8 \geq x \geq 4$); the liquid-solid phase transition of higher-order LiPSs to $Li_2S$; $Li_2S$ reversible oxidation into S. Compared with pp, AB-200 exhibits a higher peak current meaning increased active sites. In addition, AB-150 exhibits the highest peaks at 2.22 V and 1.98 V, indicating that AB-150 could improve the conversion of low-order LiPSs ($Li_2S_x$ $3 \geq x \geq 1$). In the oxidation reaction, pp exhibit a broad peak than that of the AB separator indicating a slow reaction rate. AB-200 and AB-150 are analogous peak positions at 2.39 V and 2.48 V, respectively, which are related to the decomposition of solid $Li_2S$ to soluble polysulfides and subsequently converted to solid S. It is worth noting that AB-150 has a lower onset potential (2.23 V), which indicates that AB-150 can effectively promote the $Li_2S$ decomposition. The charge-discharge curve at 90th provides more detailed information. As shown in Fig. 3b, pp, AB-200, and AB-150 batteries deliver the capacities ($Q_H$) of 208, 310, and 282 mAh $g^{-1}$, respectively, contributed by the S conversion to high-order LiPSs. it releases the capacities ($Q_L$) of 348.25, 379.75, and 650.95 mAh $g^{-1}$, respectively, originated from high-order LiPSs reduced to lower-order LiPSs that involves liquid-solid phase transition and crystal nucleation processes. As shown in Fig. 3d, the corresponding polarization voltages(ΔE) are 430, 286, and 224 mV and $Q_L/Q_H$ ratios are 1.67, 1.225, and 1.99, respectively. The initial charging stage of the battery corresponds to the solid-liquid phase transition from $Li_2S$ decomposition to high-order LiPSs. As shown in Fig. 3c, the $Li_2S$ decomposition potentials are 2.36 V, 2.32 V, and 2.23 V in pp, AB-200, and AB-150, respectively. It is worth noting that pp and AB-200 have obvious decomposition potential barriers of 33.5 mV and 33.7 mV, respectively, while AB-150 exhibits a negligible decomposition potential barrier. CV curves and discharge-charge curves show that the introduction of AB interlayer can significantly improve the utilization of $Q_H$; AB-200 cannot effectively promote the deposition and decomposition of $Li_2S$ than that of pp because of similar decomposition potential energy barriers; AB-150 can promote the uniform deposition of $Li_2S$ as evidenced by the charge deposition barrier and high $Q_L/Q_H$. As shown in Fig. 3e, pp, AB-200 and AB-150 respectively released initial capacities of 674, 783.1, 961.3 mAh $g^{-1}$ and maintained at 554.5, 717.5, and 825 mAh $g^{-1}$ at 0.2C after 100 cycles with a capacity retention rate of 82.27%, 91.36%, and 85.8%. Although AB-200 exhibits higher capacity retention than AB-150, the initial capacity is lower and the Coulombic efficiency fluctuates more dramatically. This indicates that part of the sulfur did not participate in the reaction during the initial discharge, which could be activated during cycling to release the capacity to offset the active species deactivation originating from the heterogeneous deposition of $Li_2S$ and S and the shuttle of polysulfides. Electrochemical impedance spectroscopy is used to explore the impedance information of electrodes. As shown in Fig. 3f, the high, medium and low frequency are related to the ohmic impedance ($R_s$), charge transfer impedance ($R_{ct}$), and Warburg impedance, respectively. The introduction of the AB coating has little effect on the $R_s$ of the lithium-sulfur system because of similar impedance values in pp(8.9Ω), AB-200(10.3Ω), and AB-150 (5.9Ω). $R_{ct}$ represents the Faraday reaction process, the AB-200(206.6Ω) is half of the pp (497.4Ω), which correlates with significantly improved QH. The $R_{ct}$ of AB-150(126.3Ω) is the smallest among all samples and the corresponding cell exhibits greatly improved QL. The response signal in the low-frequency region is related to ion diffusion. pp exhibits a straight line with a slope of approximately 1 corresponding to the Warburg impedance; AB-150 and AB-200 exhibit approximative straight-line slopes indicating similar ion diffusion behavior. However, the difference in the slopes of the AB coating and pp indicates that the AB coating affects the ion diffusion behavior in the electrolyte. pp, AB-200 and AB-150 cells from 0.1C to 2C are performed to simulate real-world battery usage. As shown in Fig. 4a, AB-150 releases the highest capacity at all current rates and even released a capacity of 490 mAh $g^{-1}$ at 2C which is higher than that of AB-200 (323 mAh $g^{-1}$) and pp (290 mAh $g^{-1}$) at 1C. These data suggest that AB-150 possesses a high electrochemical reaction rate, especially in the liquid-solid reaction stage, which is related to its low charge transfer impedance. However, the

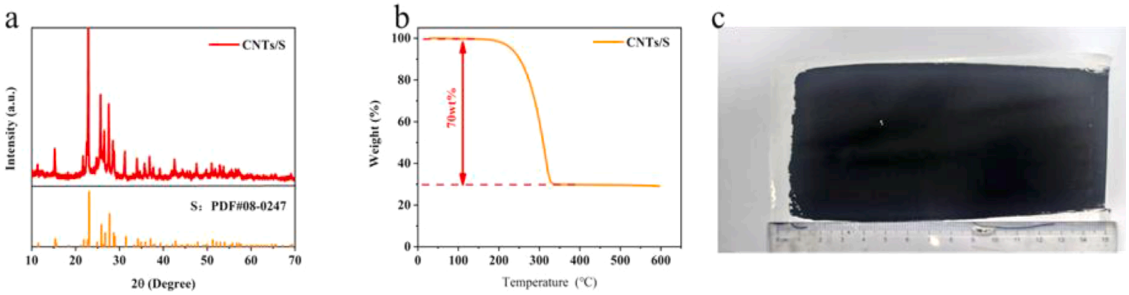



**Fig. 1.** (a) XRD pattern of CNTs/S and (b) TGA curve, (c) digital photo of AB-150 modified separator.

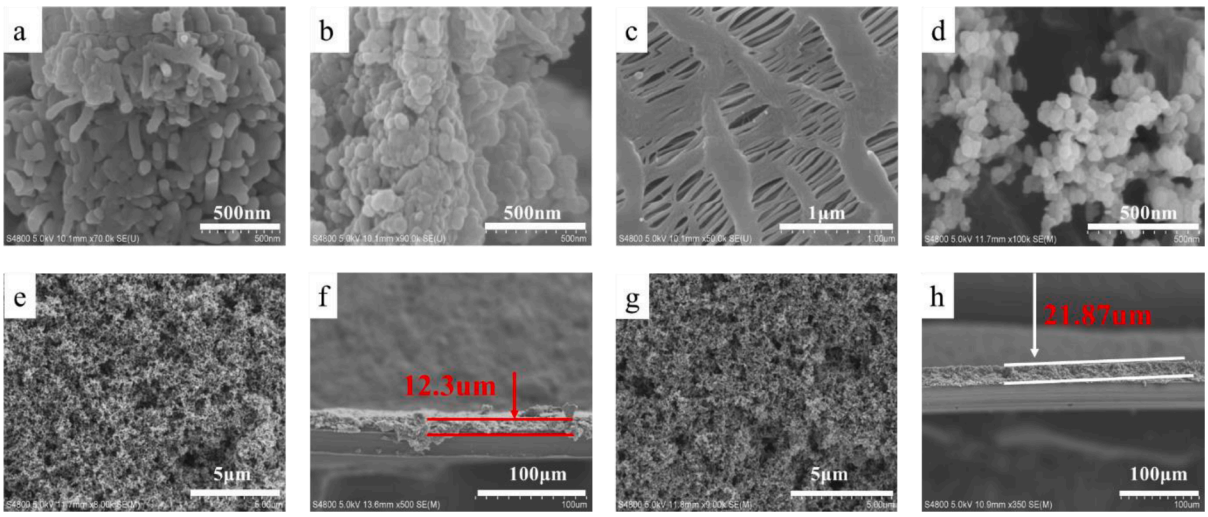



**Fig. 2.** SEM image of (a) CNTs/S powder after calcination at 155 °C; (b) CNTs/S-AB powder; (c) commercialized Celgard 2500; (d) AB powder; (e) top view and (f) cross-sectional view of AB-150 separator; (g) top view and (h) cross-sectional view of AB-200 separator.

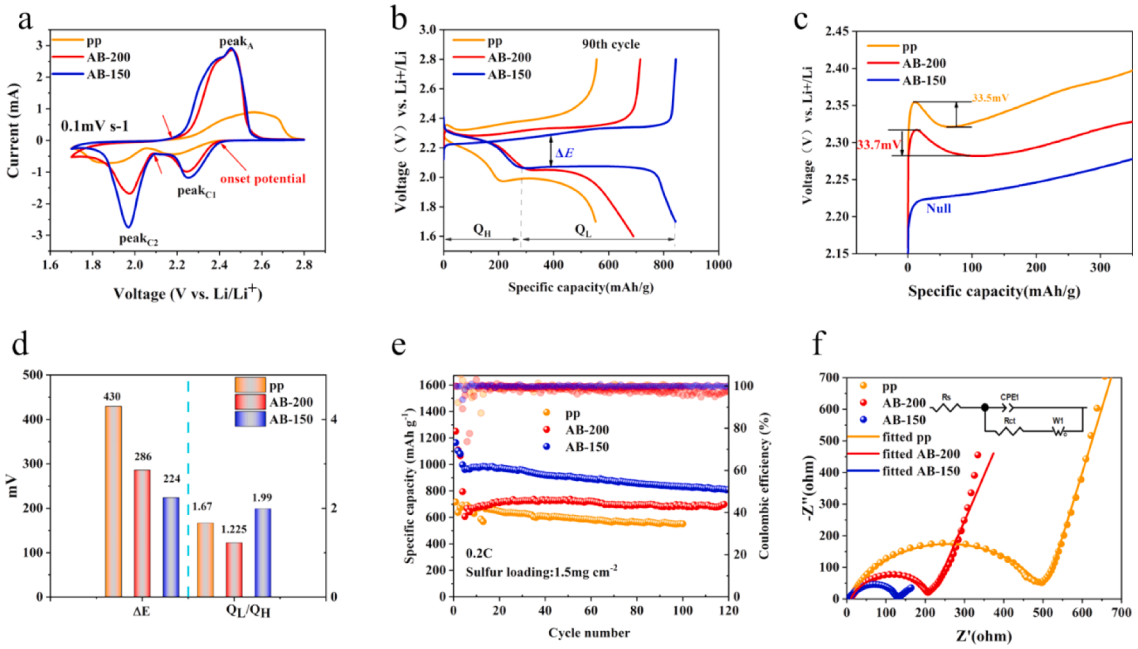



**Fig. 3.** Evaluate the electrochemical performance by various testing techniques with pp, AB-200, AB-150 separators. (a) Cyclic voltammetry curves with a scan rate of 0.1mV $s^{-1}$. (b) The charge-discharge curve of the 90th cycle at 0.2C. (c) Partially amplified charging curve of the 90th cycle at 0.2C. (d) Polarization voltage and QL/QH histogram of the battery at the 90th cycle. (e) Electrochemical performance at 0.2C. (f) EIS curve of the new battery.

reason for the reduced charge transfer impedance remains to be further explored. The charge-discharge curves of AB-150 and AB-200 are shown in Fig. 4b and c, it can be clearly seen that the two samples release similar $Q_H$ at low rate current, while $Q_L$ exhibits a huge difference at high rate current. The low-order discharge plateau of AB-200 transitions to a slowly decreasing slope whereas AB-150 exhibits a distinct voltage plateau at 1C even is still seen at 2C. Voltage-capacity curve has exhibited severe overpotential and lower QL in AB-150 at 2C because the non-polar AB has a limited effect to promote the mass transfer (polysulfides)from the cathode to the AB coating [41], although it could promote the solid-liquid reaction kinetics. The cycling stability is shown in Fig. 4d, pp, AB-200, and AB-150 release initial capacities of 445.6, 530and 934.6 mAh $g^{-1}$, the capacity maintained 158.9, 71.1, and 371.2 mAh $g^{-1}$ after 1000 cycles at 1C with a capacity decay rate of 0.064%, 0.086%, and 0.06%, respectively. It is worth noting that the pp battery completely failed at the sulfur loading of 1.5 mg $cm^{-2}$, so its sulfur loading is controlled at 0.7 mg $cm^{-2}$ in the 1C long-term cycle test. It should be pointed out that the acquired battery performance is highly competitive among the present Li−S cells literature with carbon material (Table $S4^{S1\text{-}S7}$). lean electrolyte systems are critical for commercial applications [42,43]. As shown in Fig. S1, AB-150 released initial capacities of 1055.3 and 1145.8 mAh $g^{-1}$ and maintained 958.5 and 943 mAh $g^{-1}$ after 25 cycles with E/S ratio of 10 μL $mg^{-1}$ and 20 μL $mg^{-1}$ at 0.2C. Decreasing the E/S ratio from 20 to 10, the initial discharge capacity decreased but the capacity retention increased because less electrolyte hindered the conversion of sulfur to polysulfides but this in turn mitigated the loss of polysulfides. To deliver high capacity, the AB-150 battery with an areal loading of 3.96 mg $cm^{-2}$, and lean electrolytes (8.9 μL $mg^{-1}$) were tested at 0.1C. As shown in Fig. 4e, AB-150 delivered an initial specific capacity of 656.4 mAh $g^{-1}$ and gradually

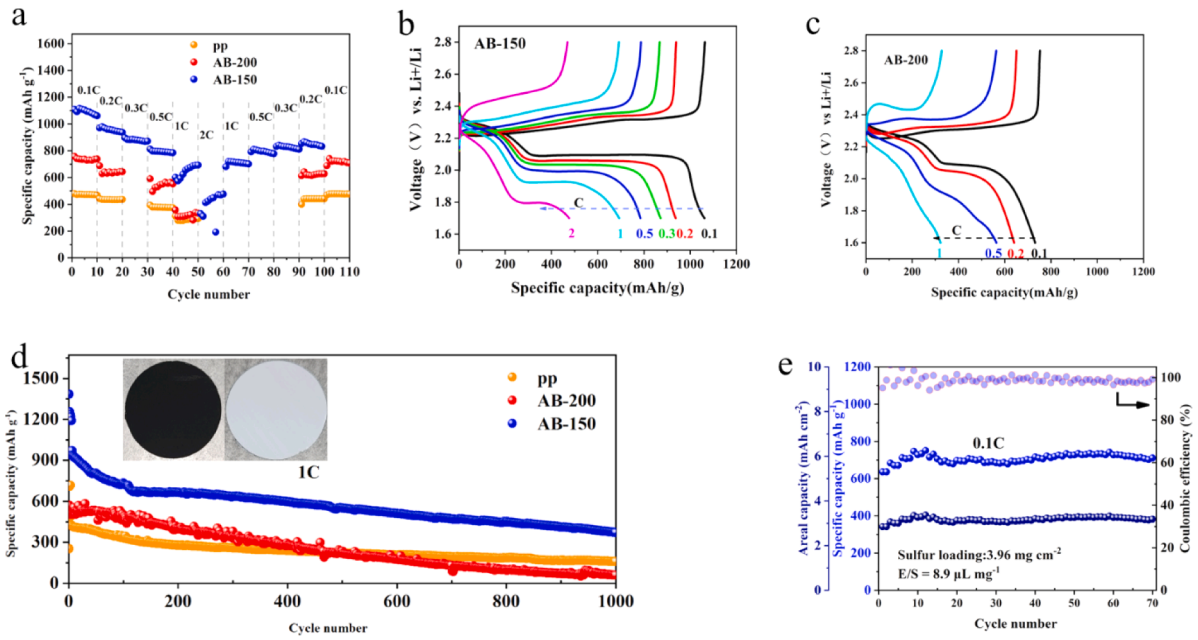



**Fig. 4.** (a) Rate capacity of pp, AB-200, AB-150. Areas without dots at the corresponding rate in PP and AB-200 represent that the corresponding current rate tests were not performed. (b) AB-150 and (c) AB-200 charge-discharge curves at different rates. (d) Cyclic stability curves of pp (sulfur loading 0.7 mg $cm^{-2}$), AB-200 (sulfur loading 1.5 mg $cm^{-2}$), AB-150 (sulfur loading 1.5 mg $cm^{-2}$) at 1C. (e) Electrochemical performance of AB-150 at 0.1C.

increased to 750.7 mAh $g^{-1}$ after the 12th cycle, which was mainly attributed to the gradual immersion of the electrolyte to activate the sulfur near the collector. AB-150 battery released a capacity of 710 mAh $g^{-1}$ with a capacity retention rate of 94.5% after 60 cycles, corresponding to an average area capacity of 3.0 mAh $cm^{-2}$.

The function of the AB coating was tested by $Li_2S_6$ static barrier experiments. As shown in Fig. 5a–e, on the right side of the H-shaped bottle, the transparent electrolyte gradually turns orange-yellow because the high concentration of $Li_2S_6$ spontaneously passes through the separator to the low concentration region, which is one of the main factors in the polysulfide deactivation. A slight yellowish color was observed at the bottom of the bottle in pp, AB-150, AB-200 after standing for 1h. AB-150 and AB-200 exhibited lighter colors compared to pp after 4 h and 12 h, indicating the ability of the AB coating to block polysulfides. The similar color of AB-150 and AB-200 after 22 h means that the two interlayers have almost the same ability to statically block polysulfides. limited block polysulfide diffusion because of the weak interaction between non-polar carbon and polysulfides.

The static adsorption test cannot simulate the dynamic electrochemical reaction process, so a $Li_2S_6$ symmetric battery was performed to explore the source of the boosted QH during discharge [44]. As shown in Fig. 6a, both CV curves exhibit strong currents response and similar shapes, which are related to the conversion of physisorbed $Li_2S_6$ to $Li_2S_4$ by the AB coating. Compared with AB-200 (760 mV, 14.5 mA), AB-150 has lower overpotential and higher peak current (490 mV, 16 mA) meaning more active sites and smaller polarization in liquid-liquid

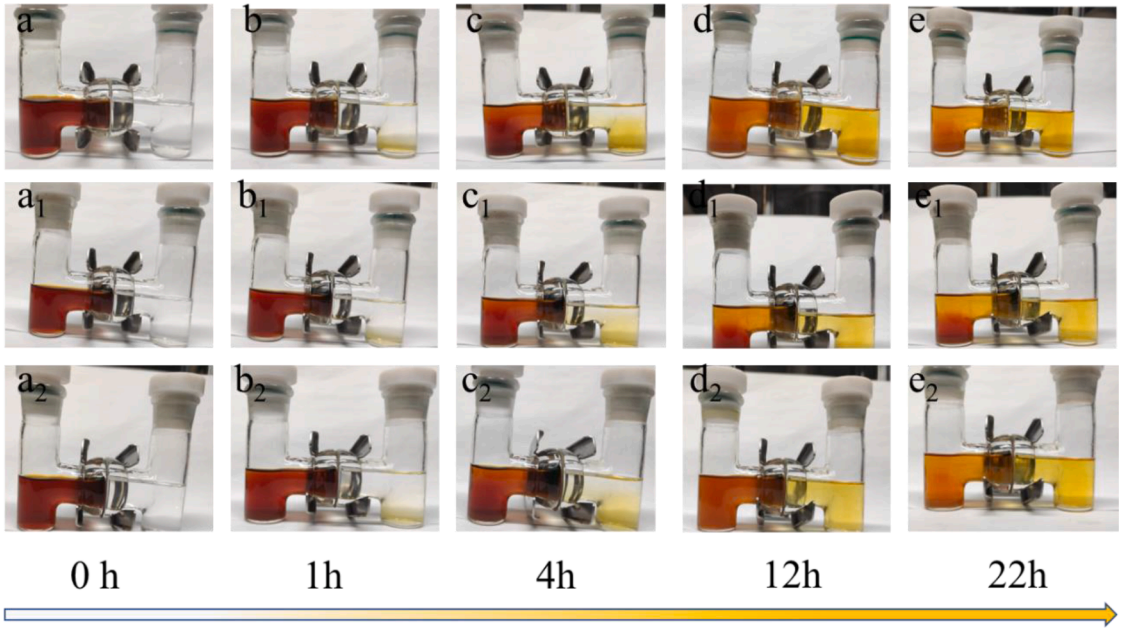



**Fig. 5.** Digitized photo of $Li_2S_6$ barrier test with separators of (a-e) pp, ($a_1$-$e_1$) AB-150, and ($a_2$-$e_2$) AB-200.

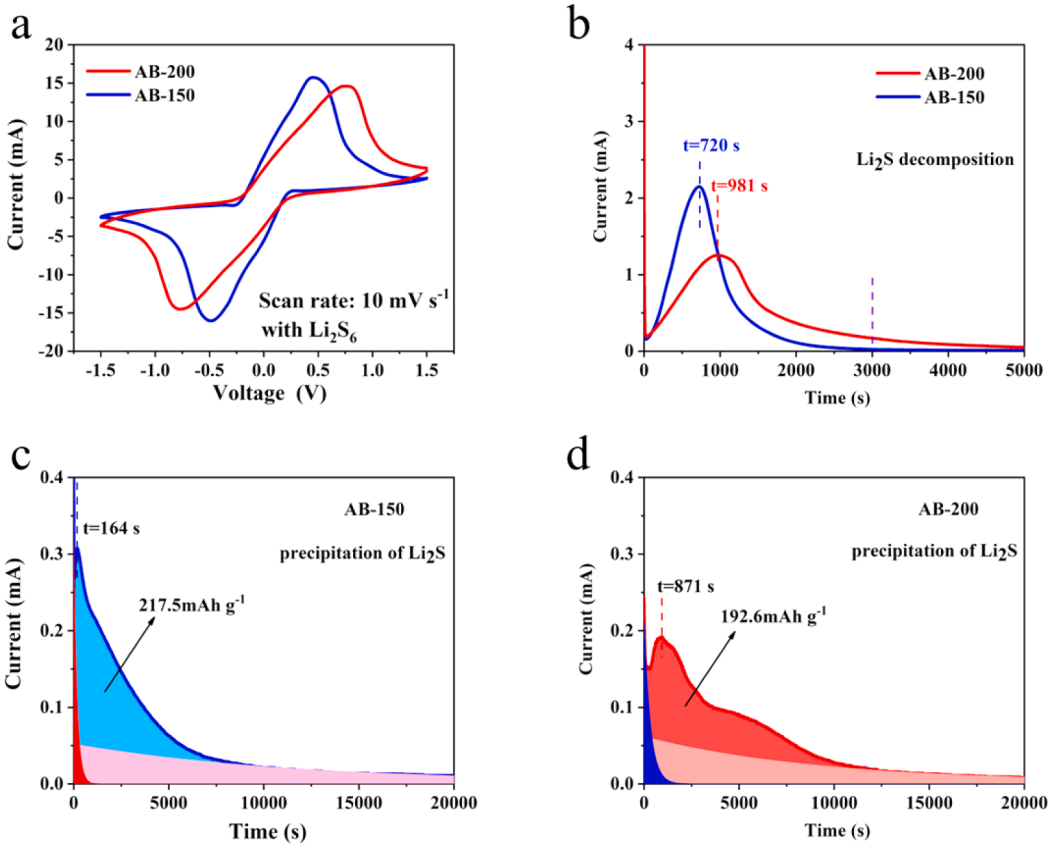



**Fig. 6.** (a) $Li_2S_6$ symmetric cell, (b) $Li_2S$ decomposition and (c and d) precipitation with AB-150 and AB-200.

conversion. This difference is more obvious at high current densities, such as the charge-discharge curves in Fig. 4b, c. symmetric cells delivering similar capacities with AB interlayers shown that AB-200 and AB-150 have similar liquid-to-liquid conversion capabilities at low current densities, consistent with the results in Fig. 3b. Static barrier experiments of $Li_2S_6$ and symmetric cell experiments have shown that the polysulfides can be immersed into AB barrier layer and continuously converted to $Li_2S_4$. We performed $Li_2S$ precipitation tests to evaluate $Li_2S$ nucleation and growth. As shown in Fig. 6c, d, the nucleation time of $Li_2S$ in AB-150 (164 s) is 5 times earlier than that of AB-200 (871 s), and AB-150 has a higher peak current (0.31 mA) compared with AB-200 (0.19 mA), indicating a faster $Li_2S$ nucleation rate and more nucleation sites. The liquid-solid conversion capacity is determined according to Faraday's law. The $Li_2S$ deposition capacity is larger in AB-150 (217.5 mAh $g^{-1}$) than in AB-200 (192.6 mAh $g^{-1}$), demonstrating that AB-150 can effectively reduce overpotential and promote $Li_2S$ deposition. AB-200 shows a broad peak at 5000 s, which is related to the slow growth of $Li_2S$ on the surface of the initially formed $Li_2S$ crystal. $Li_2S$ decomposition kinetics were tested with a similar protocol. As shown in Fig. 6b, AB-150 has the highest peak current and the shortest peak current time (2.14 mA, 720 s) than that of AB-200 (1.25 mA, 981 s). In addition, AB-200 still delivered a significant decomposition current after 3000 s, which is related to the slow decomposition of $Li_2S$. These results indicate that AB-150 can regulate the precipitation and decomposition of $Li_2S$ to promote the solid-liquid transition during charge and discharge.

The Galvanostatic intermittence titration techniques (GITT) technique was used to probe the impedance and Li-ion diffusion changes of the three samples during discharge [26,27], which are related to electrochemical reaction kinetics. $Li^+$ diffusion is calculated based on the following equation:

$$D = \frac{4}{\pi \times \tau} \times \left(\frac{m}{S \times \rho}\right)^2 \times \left(\frac{\Delta V_s}{\Delta V_\tau}\right)^2 \quad (1)$$

Where D is the lithium-ion diffusion coefficient ($cm^2\ s^{-1}$), τ is the discharge pulse time (s), m is the mass of sulfur in the positive electrode (g), ρ is the density of sulfur (2.36 g $cm^{-3}$), and S is the area of positive electrode ($cm^2$), $\Delta V_s$ is the steady-state potential change through the current pulse, $\Delta V_\tau$ is the potential difference during the galvanostatic pulse after eliminating the IR drop, the meaning of these symbols is also marked in Fig. 7a. As shown in Fig. 7b, when the depth of discharge is about 25%, AB-200 exhibits higher Li-ion Diffusion ($1.028 \times 10^{-10} cm^2 s^{-1}$) compared with pp ($2.322 \times 10^{-11} cm^2 s^{-1}$). AB-150 ($1.124 \times 10^{-10} cm^2 s^{-1}$) with ion diffusion similar to AB-200 corresponding cells released similar $Q_H$ capacity. It represents the conversion of low-order LiPSs to $Li_2S$ at a depth of discharge of about 76%, which is the main contributing stage to the capacity of Li-S batteries. As shown in the intercalation image in Fig. 5b, pp, AB-200, AB-150 have $Li^+$ diffusion of $2.253 \times 10^{-13} cm^2 s^{-1}$, $2.500 \times 10^{-13} cm^2 s^{-1}$, $and 6.87710^{-13} cm^2 s^{-1}$, respectively. These results show that introducing AB coating into Li-S batteries can change lithium-ion diffusion ability and improve $Q_H$while the obvious improvement in QL is attributed to the high Li-ion diffusivity in AB-150. The internal resistance change of the battery is explored by the GITT technique. As shown in Fig. 7c, pp has an impedance of 112.5-157.78Ω, which is significantly higher than that of AB-200 (29.26-44.25Ω) and AB-150 (24.7-29.25Ω). It is worth noting that during discharge, the impedances of pp and AB-200 exhibit larger fluctuations of 45.28Ω and 14.99Ω than that of AB-150 (4.55Ω). These test dates indicate that AB-150 possesses the smallest impedance fluctuation during discharge and releases the highest QL, which is associated with the deposition of insulating $Li_2S$ and S at the cathode-coating interface.

After 100th cycles at 0.2C, the batteries were charged to 2.8V and

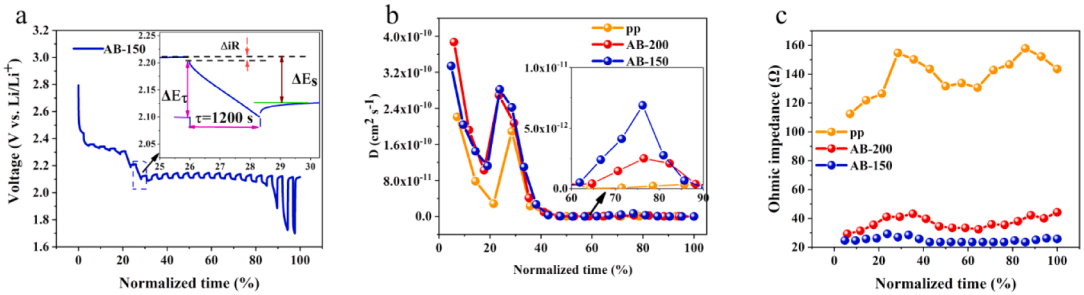



**Fig. 7.** (a) GITT curve of AB-150. (b) Li-ion diffusion coefficient of pp, AB-200, and AB-150 battery that calculated based on GITT curve. (c) impedance versus depth of discharge curve.

then disassembled to explore the change in the morphology of the AB coating. No large bulk aggregation is observed for AB-150 (Fig. 8a), the magnification image (Fig. 8b) shows a spherical structure but the particle surface and space are uniformly filled. Surprisingly, however, the surface of AB-200 (Fig. 8c) has a large number of uniformly distributed fusiform structure with a length of about 500 nm, which are rarely observed in other's research. The magnification image (Fig. 8d) shows few AB were observed out of the sheet-like structure. These pictures show that AB-150 can regulate the uniform nucleation and deposition of solid-phase particles, which is the main factor for small impedance fluctuation, small charge transfer impedance, and strong lithium-ion diffusion ability. The composition of the surface deposits is analyzed by EDS spectroscopy. As shown in Fig. 8g, AB-150 and AB-200 exhibit the same peak positions indicating that the deposits have similar compositions containing C, O, F, and S elements. The F and S elements are mainly derived from the electrolyte (LiTFSI) and the element S The corresponding element relative contents are shown in Fig. 8h, the C content in AB-150 is 47.7 wt.% much higher than that in AB-200 (20.54 wt.%), which indicates that the surface of AB-200 is heavily covered by the deposition layer; The F content in AB-200 (16.83 wt.%) is more than three times that in AB-150 (5.5 wt.%), which indicates that the electrolyte in AB-200 is abundantly present on the surface of the AB interlayer. The S content in AB-150 (27.95 wt.%) and the sulfur content in AB-200 (27.18 wt.%), indicate that the AB coating can effectively physically block the loss of active materials.

The morphology of Li deposition was further explored by SEM images. As shown in Fig. 9a–d, the fresh Li surface is smooth. After 50 cycles, the most severe particle agglomeration and the roughest surface were observed with pp separators, while the distance between the deposited particles was larger with AB-200 than that with AB-150. As High-resolution SEM images in Fig. 9f–h, the Li precipitates are irregular with pp separator. It is significantly different in morphologies of lithium dendrite with both AB coatings. it is made of spherical particles tightly packed with a uniform size of around 500 nm with an AB-150 separator, and a blockbuster-like-mossy layer contacted with spherical particles with the AB-200 separator [45]. These results indicated that AB coating can regulate the growth of Li dendrites and the formation of dead Li [46]. As shown in Fig. 9i, for precipitate, the intensity of sulfur element with the pp membrane is ten times higher than with the AB coating and is slightly lower with the AB-150 membrane compared to the AB-200, indicating that AB-150 has the strongest ability to limit the corrosion of lithium anodes by polysulfides.

## 4. Conclusions

In summary, we utilize AB as a modified separator for lithium-sulfur batteries. Electrochemical tests show that AB coating can effectively improve $Q_H$, and more importantly, AB-150 further improves $Q_L$, high rate, and long cycle performance. CV, EIS, and GITT tests showed that AB-150 improved the reaction kinetics, lithium-ion diffusion ability, electrode interface impedance, and charge transfer impedance of low-order polysulfides. Cell dissection after cycling found that despite the presence of similar elements on the surfaces of AB-150 and AB-200, there were distinct differences in how the solid particles were deposited. During the liquid-solid reaction, the nucleation and growth of crystals are regulated through the AB coating, which is the fundamental reason for the improved electrochemical performance. These understandings can also be used to design novel materials to tune the nucleation and uniform distribution of S and $Li_2S$ crystals to further advance the commercialization of lithium-sulfur batteries.

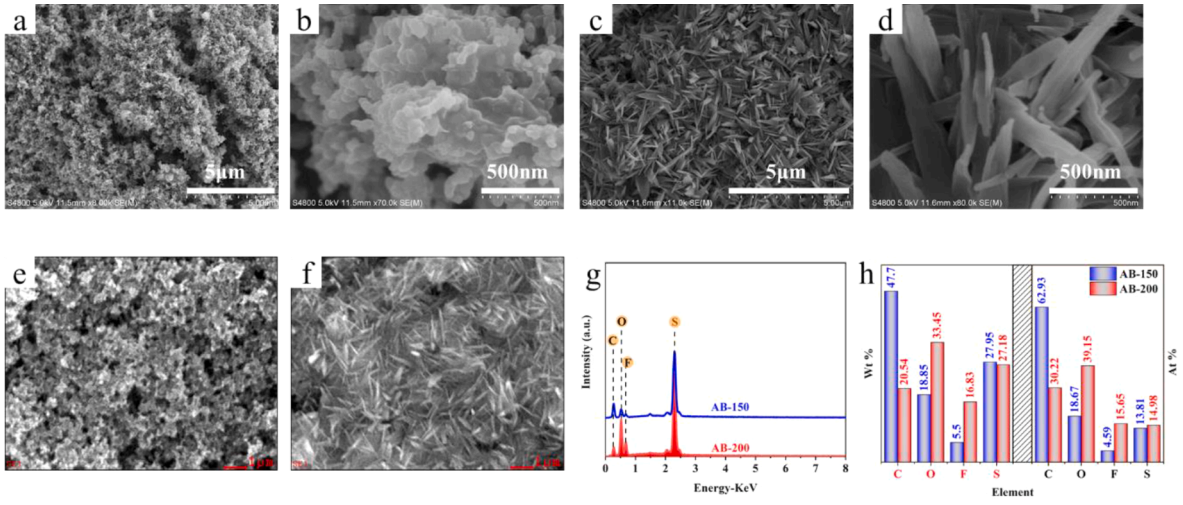



**Fig. 8.** The battery is disassembled after being charged to 2.8V after 100 cycles. SEM images of AB-150 membrane after cycling (a,b) and AB-200 membrane after cycling (c,d). EDS scanning area of AB-150 (e) and AB-200 (f), EDS (g) and relative element content (h) of AB coatings.

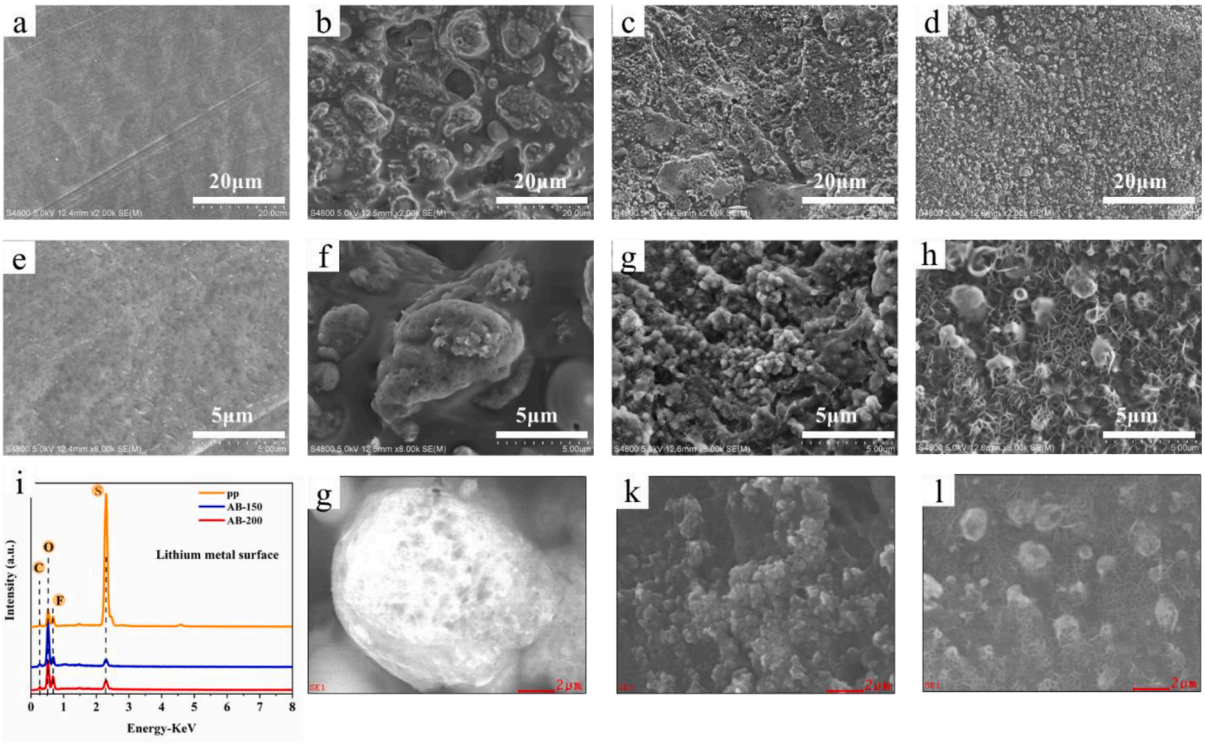



**Fig. 9.** The battery is disassembled after being charged to 2.8V after 50 cycles. SEM images of fresh Li surface (a, e) and with pp (b, f), AB-150 (c, g), AB-200 (d, h) separators. Scanning area of pp (g), AB-150 (k), and AB-200 (l) and corresponding EDS (i).

## CRediT authorship contribution statement

**Xin-Liang Men:** Conceptualization, Methodology, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Teng Deng:** Visualization, Data curation, Validation. **Xue-Chao Jiao:** Supervision, Software. **Lai-Tao Qu:** Investigation. **Kai-Ning Wen:** Investigation. **Jiang-Xuan Che:** Software. **Li-Ping Chen:** Resources, Writing – review & editing. **Sen Li:** Data curation. **Juan Wang:** Supervision, Project administration, Funding acquisition.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.electacta.2022.141100.

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